



Banana Bawp It!

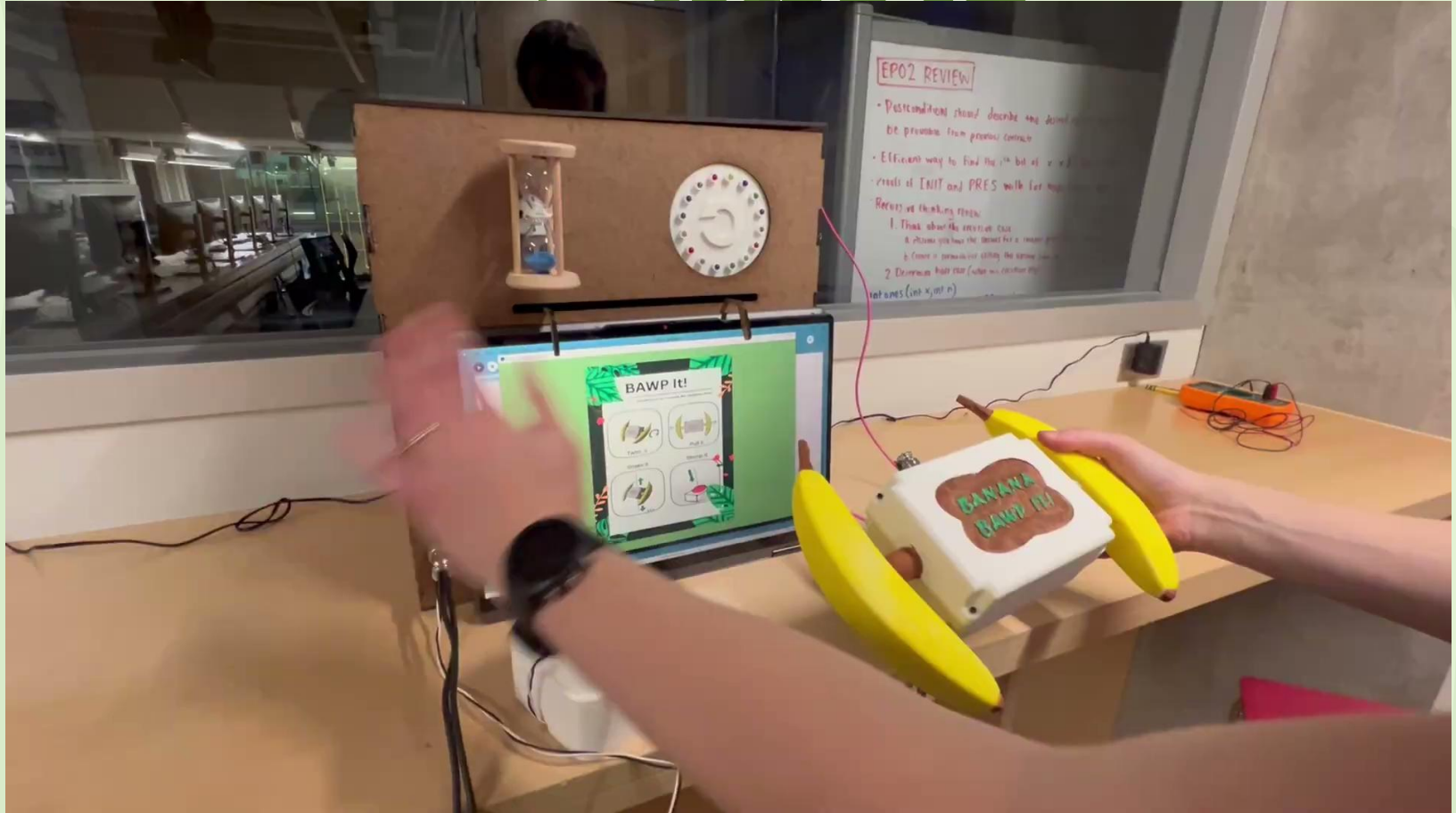
Video Presentation

Brought to you by Team 1
Hopewell Feldmann, Min Seo Kim, Katherene Qi, Arnav Srikanth





Demonstration

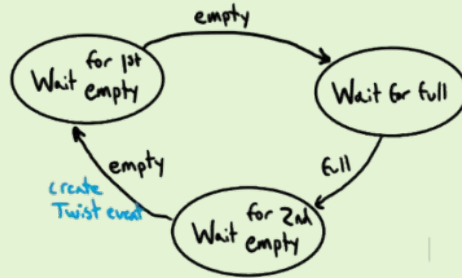


Details of Implementation



3 State Machines

Twist State Machine

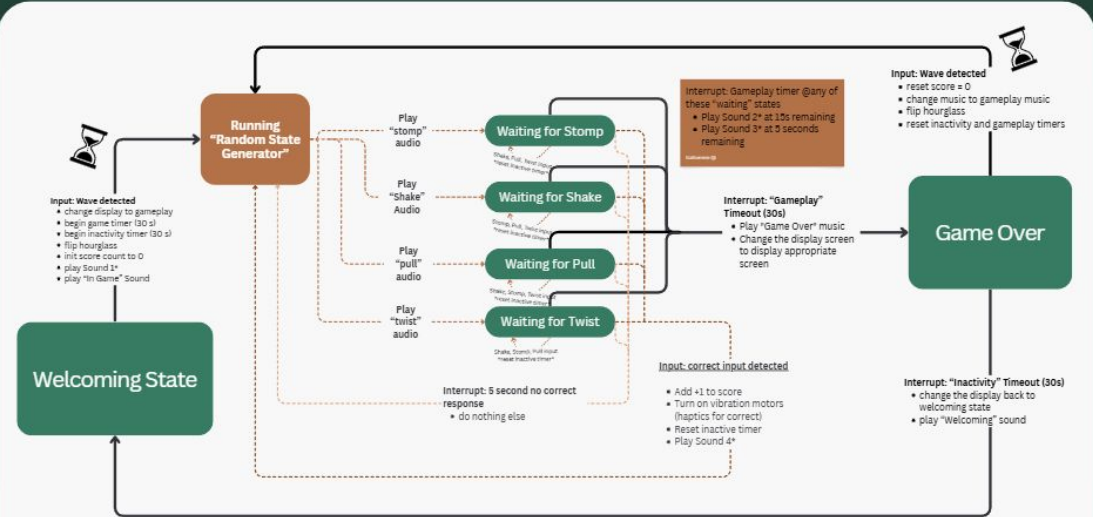


Debounce State Machine



Banana Jungle State Diagram

Team 1



In Game Sounds:

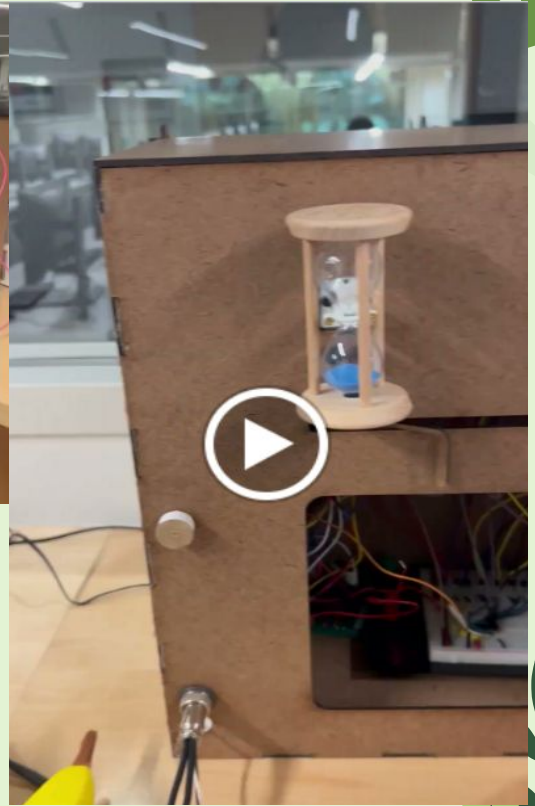
- Sound 1: "You have 30 seconds, go!"
- Sound 2: "15 seconds left"
- Sound 3: "5 seconds left"
- Sound 4: Success ping

Sounds in States:

- Welcoming: "Donkey Kong Country - Cranky's Theme"
- In Game: "Donkey Kong Country - Simian Segue"
- Game Over: "Donkey Kong Country - Cranky's Theme"

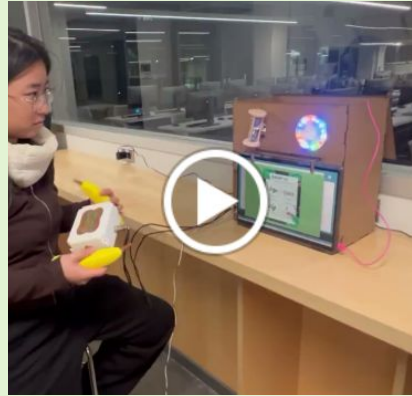
Gameplay Add-ons

- Sound Elements
- Timers
 - Inactive
 - Gameplay
- LED array
- Hourglass Timer



Controller Features

- Controller - 3 inputs
- Stomp button
- Vibration Motors

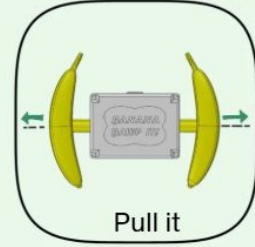


BAWP It!

Brought to you by: Hopewell, Min, Katherine, Arnav

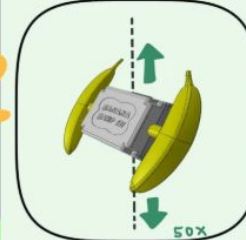


Twist it

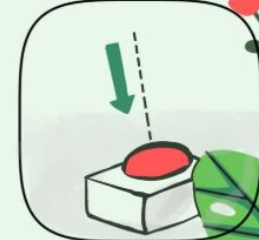


Pull it

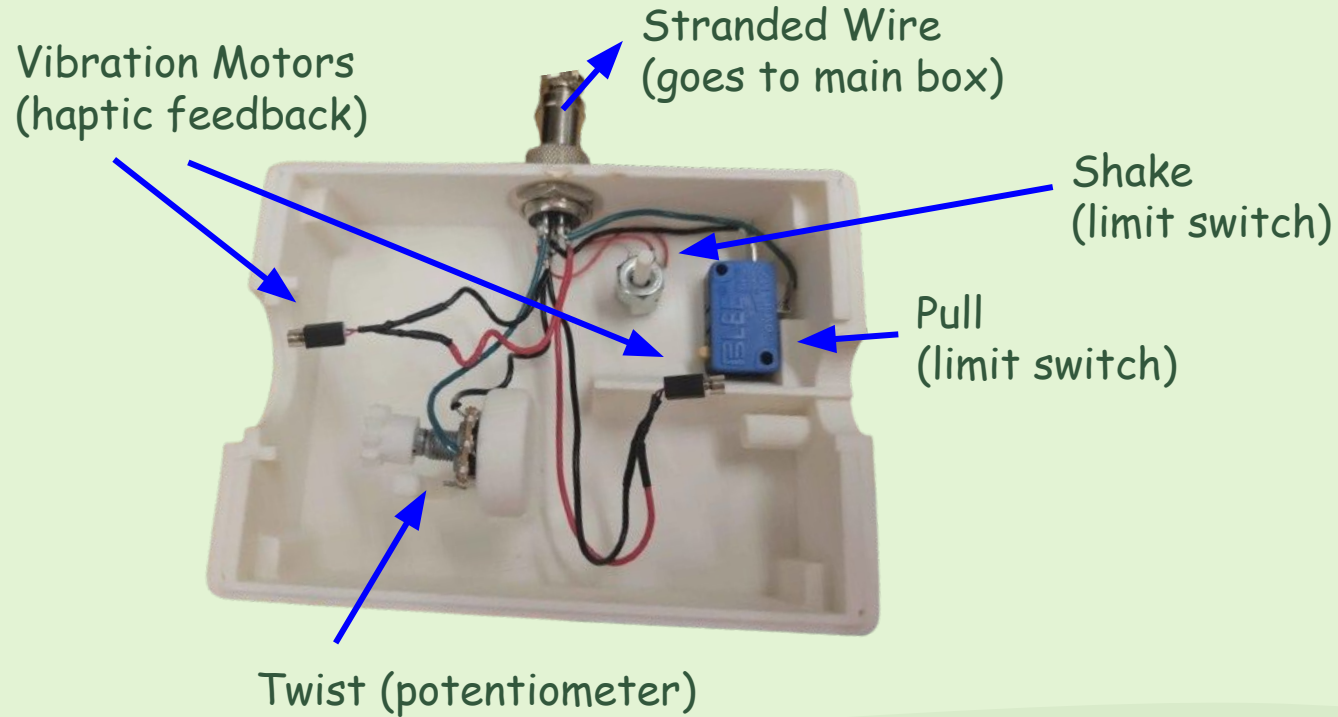
Shake it



Stomp it



Controller Hardware



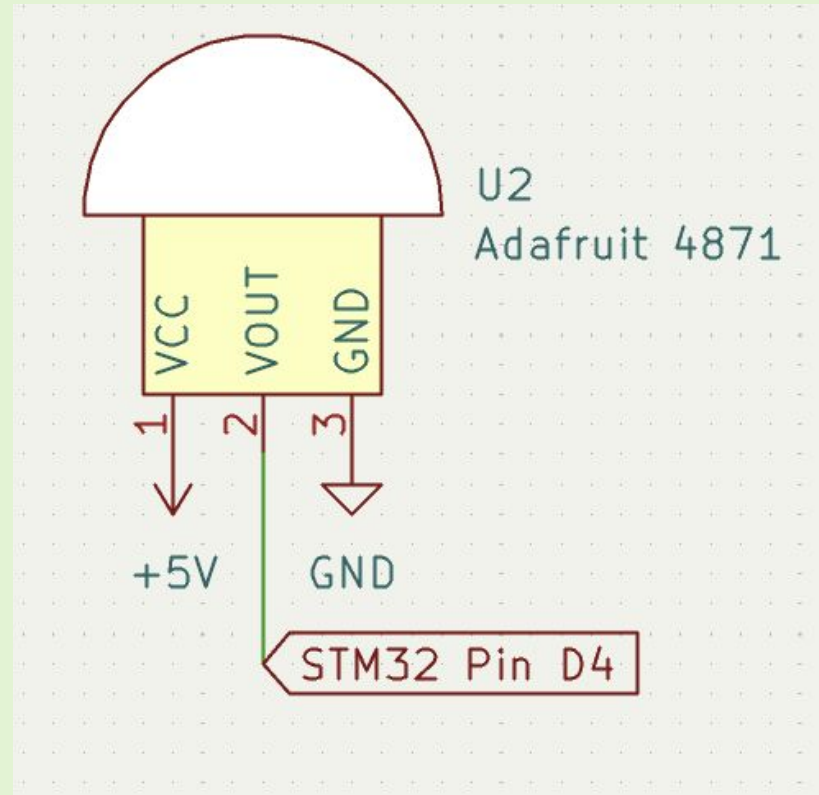
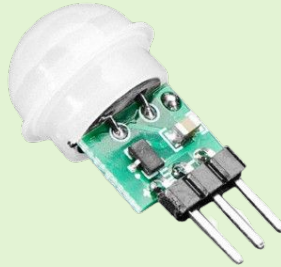
Display

- Computer Reads Microcontroller readings
- Processor (the app)
 - Plays sounds
 - Control the Display



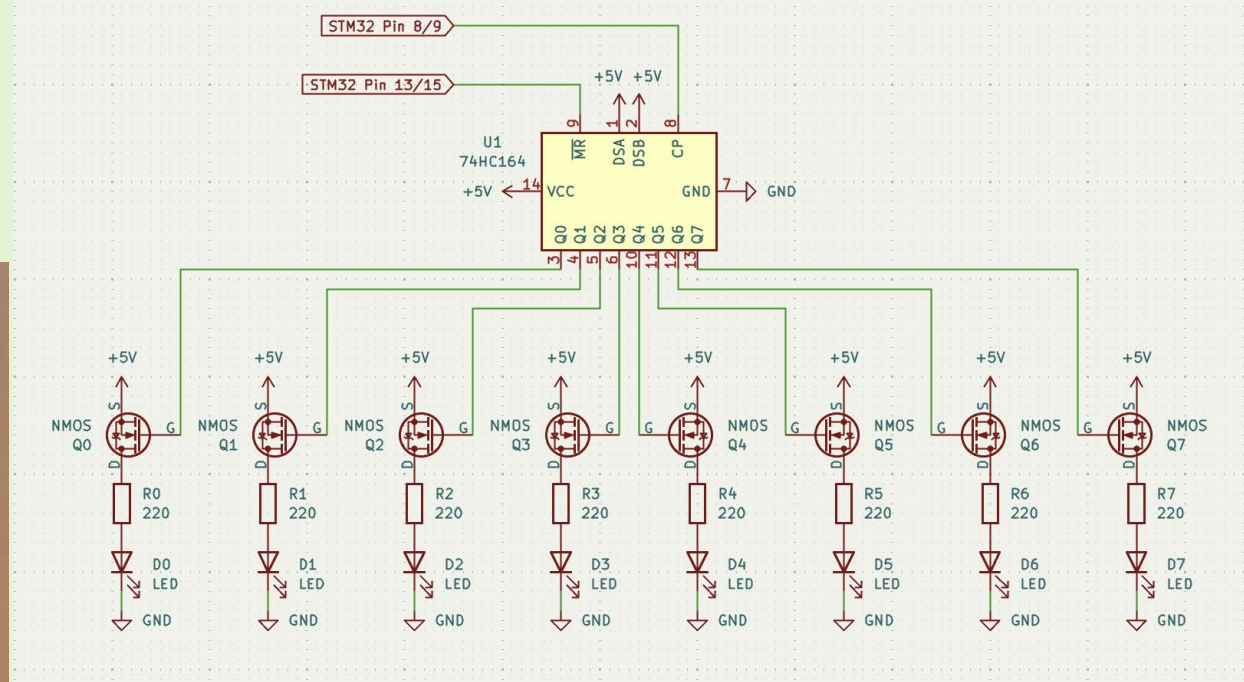
Motion Sensor

- Detect rising edge to begin gameplay



LED Array

- Analog Input + Output

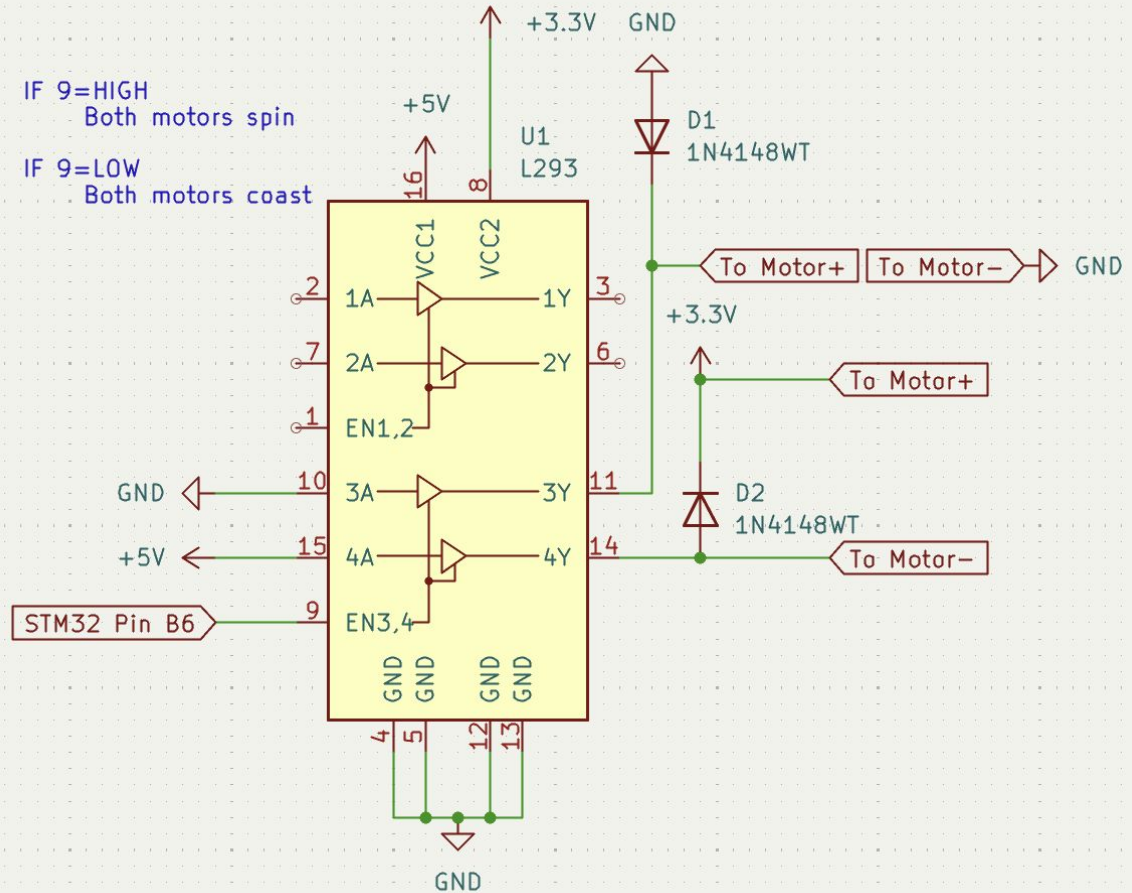


Vibration Motors

- Unidirectional driver

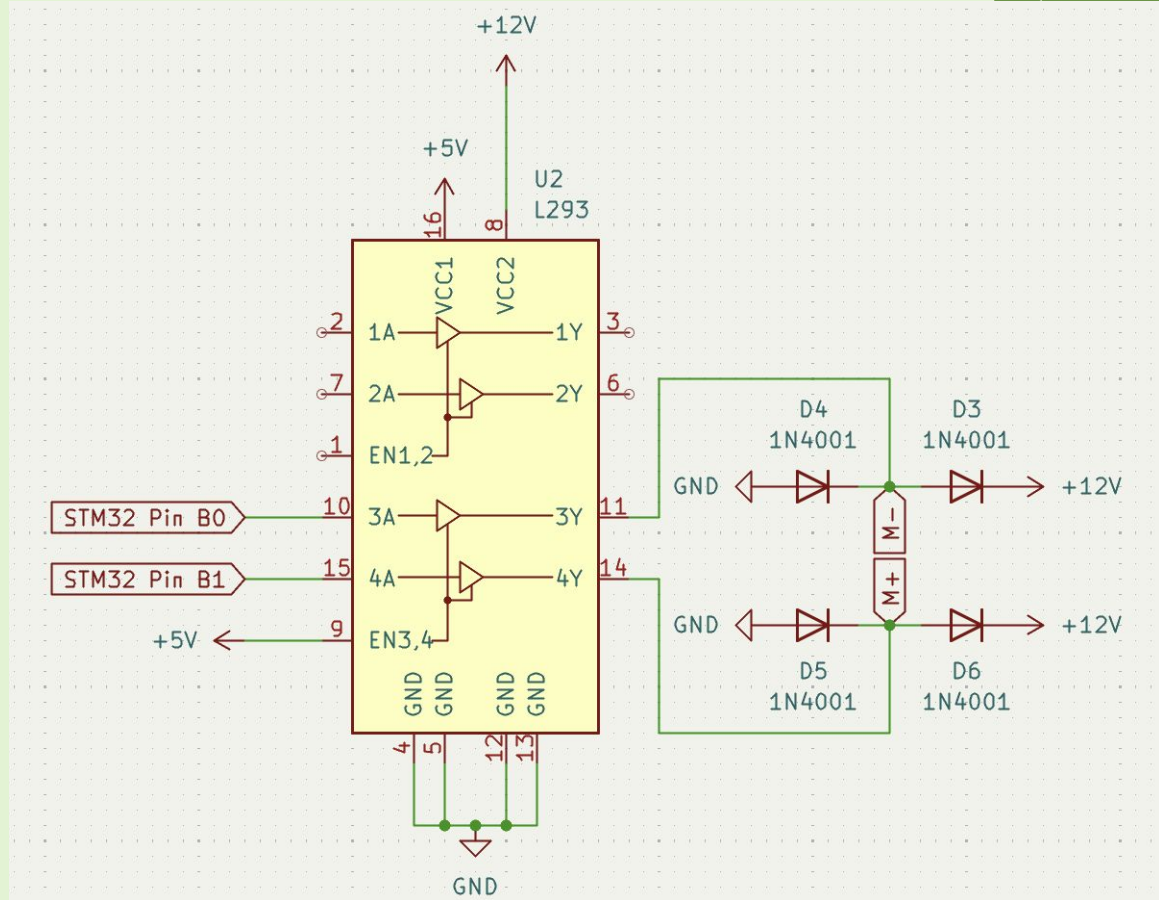


IF 9=HIGH
Both motors spin
IF 9=LOW
Both motors coast



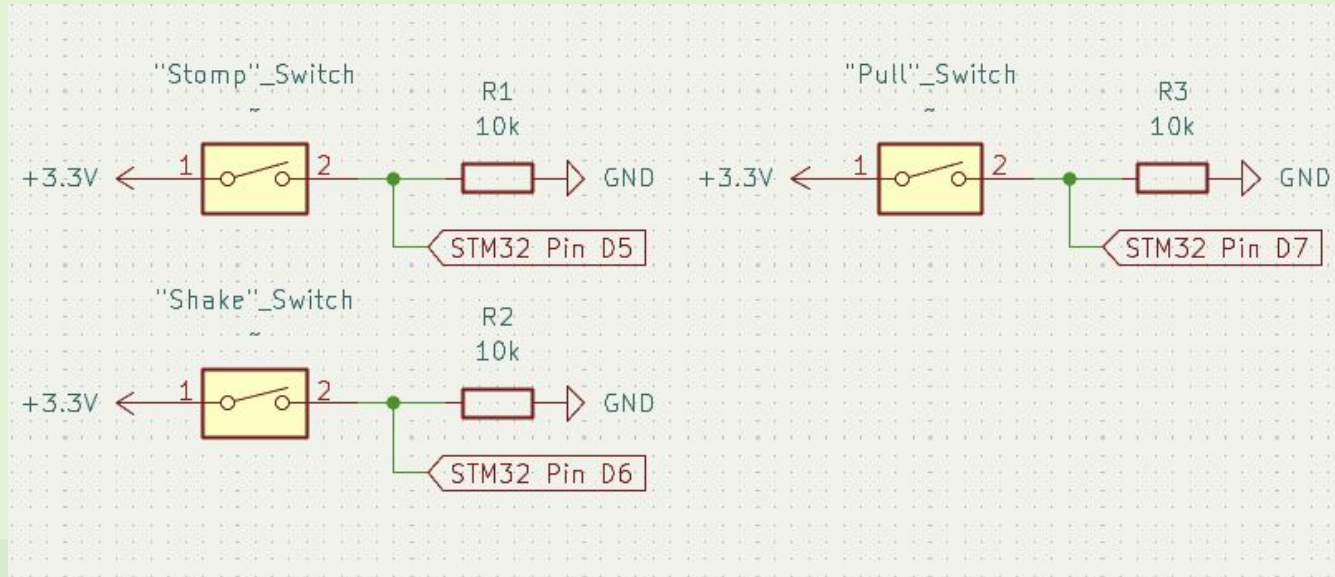
Hourglass

- PID Motor Control



Limit Switch

- Pull, stomp, shake tasks



Final Statements



Lessons Learned + Wisdom for Future Students

Hardware:

1. Use much lower infill for printing, assume each iteration won't be final
2. For smaller gears, use larger module so gears don't slip
3. Make electronics accessible for hardware debugging

Software:

1. Leave plenty of time for software debugging
2. Plan out pins, timers, and interrupts before for better organization



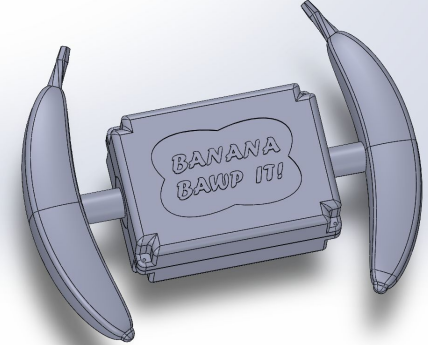
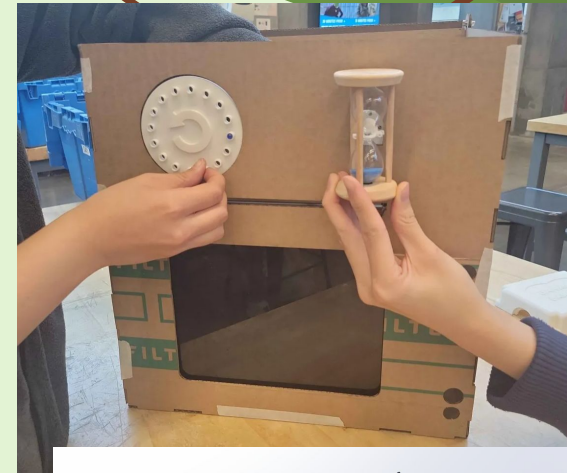
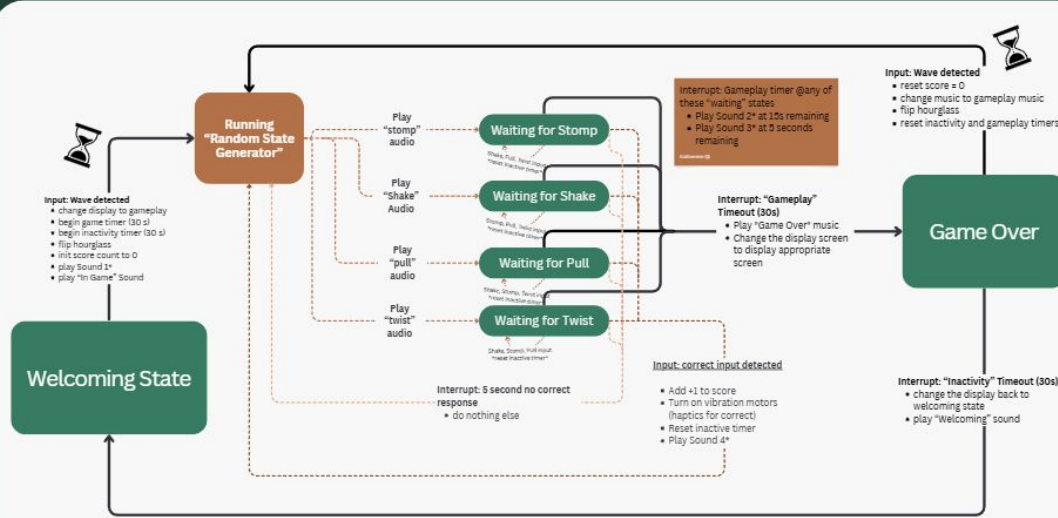
**Thank You
for Watching!**

Implementation between Inputs and State Machine



Banana Jungle State Diagram

Team 1



In Game Sounds:

Sound 1: "You have 30 seconds, go!"

Sound 2: "15 seconds left"

Sound 3: "5 seconds left"

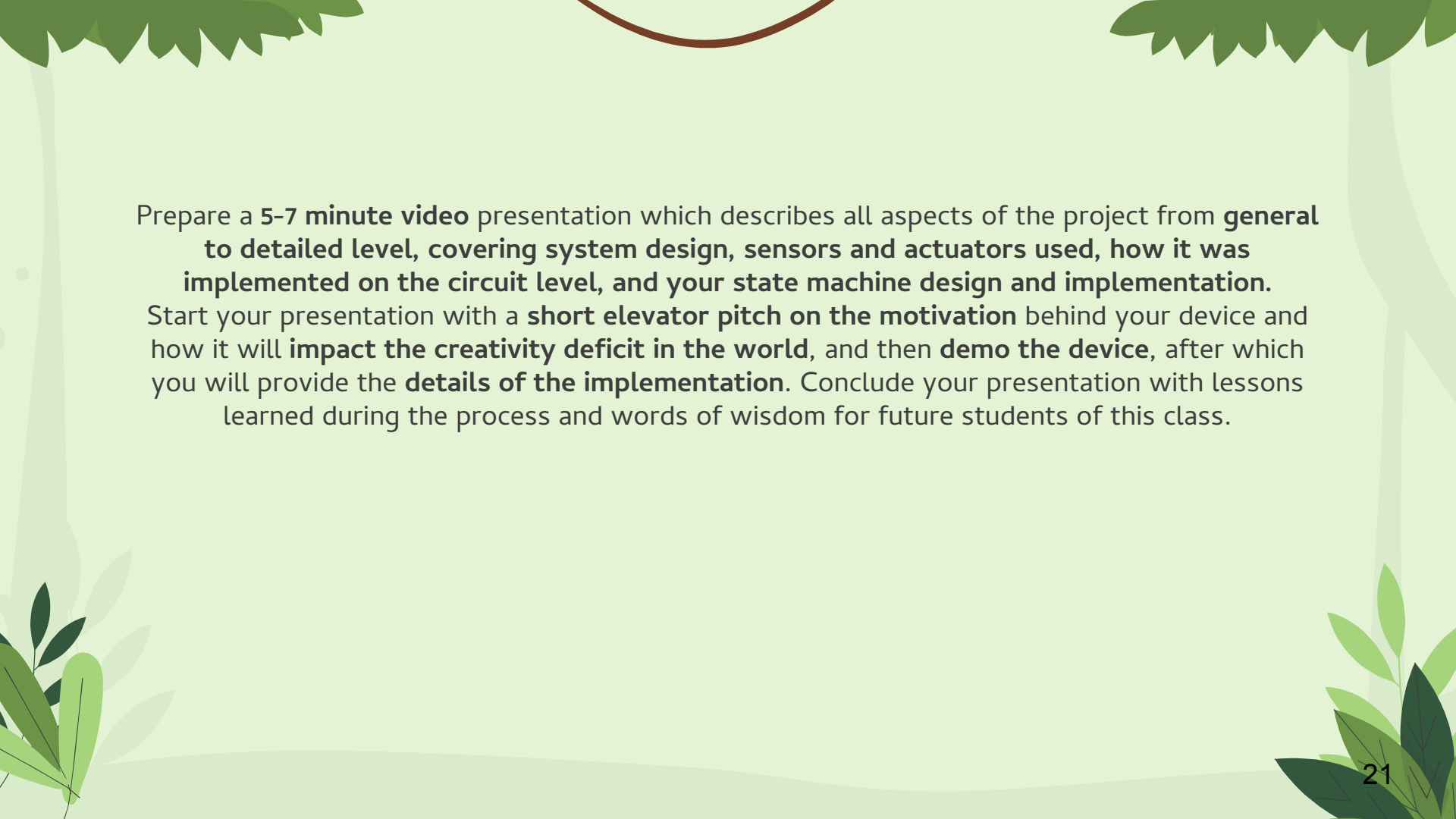
Sound 4: Success ping

Sounds in States:

Welcoming: "Donkey Kong Country - Cranky's Theme"

In Game: "Donkey Kong Country - Simian Segue"

Game Over: "Donkey Kong Country - Cranky's Theme"



Prepare a **5-7 minute video** presentation which describes all aspects of the project from **general to detailed level, covering system design, sensors and actuators used, how it was implemented on the circuit level, and your state machine design and implementation.** Start your presentation with a **short elevator pitch on the motivation** behind your device and how it will **impact the creativity deficit in the world**, and then **demo the device**, after which you will provide the **details of the implementation**. Conclude your presentation with lessons learned during the process and words of wisdom for future students of this class.

1. Banana game: to bring more banana themed gameplay to the world/the market
2. System Design:
 - a. Hourglass motor (creative display of time)
 - b. LED array (Analog input)
 - c. Vibration Motors (haptic feedback)
 - d. Motion Sensors (exits Welcoming)
 - e. Limit Switches (2) (2 silly interactions)

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